

Telltale Tails: A New Approach to Estimating Unique Market Information Shares

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Abstract

The trading of securities on multiple markets raises the question of each market's share in the discovery of the informationally efficient price. We exploit salient distributional features of multivariate financial price processes to uniquely determine these contributions, thereby resolving the main drawback of the widely used Hasbrouck (1995) methodology, which merely provides upper and lower bounds of a market's information share. We show how tail dependence of price changes, which may emerge as a result of differences in market design, can be exploited to estimate unique information shares. Two empirical applications illustrate the practical use of the new methodology.

I. Introduction

One of the most frequently asked questions in empirical finance is “Where is the market?” Whether in the case of cross-listings of stocks or newly developed derivative markets, this question has stirred up an enormous amount of research. Booth, Lin, Martikainen, and Tse (2002), for instance, examine the role of upstairs and downstairs markets in the price discovery process at the Helsinki Stock Exchange, while Huang (2002) estimates the contributions of market makers and electronic crossing networks to the price formation of NASDAQ stocks. Hasbrouck (2003) analyzes the importance of different trading venues for the price discovery process of U.S. equity indexes. The share of the futures market in U.S. Treasury price discovery is the focus of a study by Mizrahi and Neely (2008), and Blanco, Brennan, and Marsh (2005) estimate the share of the bond

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market and the market of credit default swaps (CDSs) in the process of pricing credit risk. Although they deal with different research questions in different trading environments, all these studies apply Hasbrouck's (1995) information shares methodology, the prevalent approach to measuring contributions to price discovery.

In this paper we resolve the main drawback of Hasbrouck's (1995) methodology, which does not provide a unique measure but merely upper and lower bounds of a market's information share. These bounds can diverge considerably and hinder detecting the market leading price discovery. Our approach identifies a unique information share by exploiting a salient distributional property of financial markets data, tail dependence. Tail dependence describes the phenomenon whereby the correlation between financial returns in the tails differs from that in the center of the distribution (see also Longin and Solnik (2001)). By utilizing this feature, we are able to deliver a more accurate measure that can be applied to study price discovery in various fields of financial research.

Information shares are defined as each market's contribution to the variance of the efficient price innovations. However, this variance cannot be decomposed without further restrictions when the observable market price innovations are contemporaneously correlated. The correlation increases with longer sampling intervals as the ordering of price change events in time becomes blurred. In applications like those considered in the present paper, market prices are available at most on a daily frequency. Shortening the sampling interval in order to mitigate the contemporaneous correlation and to retain the Granger-causal ordering of events is not an option. And even if it was possible, there may be good reasons to refrain from doing it. At very high frequencies, data errors like asynchronous clocks across markets or reporting delays could severely bias the results. It is this trade-off between precision and robustness that led, for example, Hupperets and Menkveld (2002), who examine price discovery for interlisted stocks, to sample prices at 5-minute intervals despite the possibility of using data at a higher frequency.

The standard solution to disentangle contemporaneously correlated price innovations when decomposing the efficient price variance is the Cholesky factorization of the covariance matrix of price innovations across markets. However, the matrix decomposition implies an informational hierarchy of markets that is hardly ever tenable. The common workaround is to permute the order of markets in the Cholesky factorization that results in upper and lower information share bounds. When these bounds diverge, which is the case when the contemporaneous correlation of market prices is strong, they measure contributions to price discovery very inaccurately. Indeed, Hupperets and Menkveld (2002) estimate lower and upper bounds that differ by up to 50 percentage points. Similarly, the information share bounds reported by Booth et al. (2002) for the upstairs and downstairs markets at the Helsinki Stock Exchange diverge by about 80 percentage points.

Our approach toward unique information shares is related to the identification of structural shocks through heteroskedasticity (see also Rigobon (2003)) and identification through nonnormal innovations (see also Lanne and Lütkepohl (2010)). Our work connects these concepts with the tail-dependence property of financial price processes. We show how tail-dependent price changes can emerge

in a microstructural model as a result of differences in liquidity and informational efficiency across markets. Using a matching reduced-form time-series representation, we explain how tail dependence of market price innovations can be exploited to identify unique information shares. The intuition behind the methodology is that the occurrence of an extreme price movement in one market can indicate either an informative event or a transitory liquidity shock. Contemporaneous price movements in the other markets can then reveal the informational content of the large price change and help to identify market-idiosyncratic innovations and, as a result, unique information shares.

Drawing the theoretical rationale from a microstructural reference model, we assume that market-idiosyncratic price innovations come from mixture distributions. This specification can account for the tail dependence that we exploit to identify information shares. The restriction on the data is that the correlations between the market price innovations in the tails and in the center of the joint distribution must be sufficiently different. We propose 2 specification tests to see whether the data possess the necessary properties.

Since there are no identifying restrictions suggested by financial theory, exploiting distributional properties for the identification of information shares is quite appealing. However, there is a hitch. We show that without further identifying restrictions, one would merely identify the *set* of information shares, but not their *allocation* to the markets. We offer a solution by proposing identifying constraints that naturally arise within the *one security—multiple markets* framework.

In the empirical section we present 2 applications of the new methodology. In the 1st application, we study the contributions of the CDS and the corporate bond market to discovering the price of credit risk. The results emphasize the informational leadership of the more liquid credit derivatives market during the pre-crisis period. They also corroborate conclusions of previous studies that identify market liquidity as the most important variable for explaining market information shares (see also Yan and Zivot (2010)). Liquidity, as a result of market design, attracts trading volume and promotes a market's leadership in price discovery. The methodology proposed in this paper systematically exploits the informational content of these market design effects to deliver a unique measure of a market's information share. Our 2nd application deals with price discovery in the futures and spot markets for European CO₂ emission allowances (EUAs). While tail dependence in price changes is pronounced in credit risk markets, it is less prevalent in the EUA spot and futures markets. We argue that in such a situation the new methodology should be applied cautiously, and we emphasize the importance of checking the validity of the identifying assumptions using the proposed model specification tests.

The remainder of the paper is organized as follows: Section II.A reviews the theoretical and statistical foundations of Hasbrouck's (1995) approach to measuring contributions to price discovery and defines some basic notation. Section II.B illustrates how tail-dependent price changes can emerge in a microstructural model, and Section II.C shows how tail dependence can be exploited to provide unique information shares. In Section II.D we resolve the identification problems associated with the computation of *tail-dependent information shares*, and in Section II.E we provide an intuitive explanation of how identification on

the basis of tail dependence works. Estimation procedures and specification tests are developed in Section II.F.1. In Sections III.A and III.B we discuss the results of the 2 empirical applications, and Section IV concludes.

II. Methodology

A. Measurement of Price Discovery across Markets: Theoretical Framework and Notation

In Hasbrouck's (1995) *one security—multiple markets* framework, identical or closely related assets are traded simultaneously on n markets. The market prices (quotes or transaction prices) for those assets prevailing at time t , $\mathbf{p}_t = (p_{1,t}, \dots, p_{n,t})'$, are assumed to incorporate a common component, the efficient price m_t , a notion that is substantiated by specifying

$$(1) \quad m_t = \lim_{J \rightarrow \infty} \mathbb{E}(p_{i,t+J} | I_t), \quad i = 1, 2, \dots, n.$$

Equation (1) renders m_t , a martingale, which is consistent with asset pricing theory, since time and state preferences can be neglected on the frequencies at which we usually sample the data for price discovery studies. The observed market prices differ from m_t due to transitory microstructure effects like bid-ask spreads, minimum tick sizes, stale quotes, or inventory management. Lehmann (2002) formalizes these ideas in a basic microstructural framework,

$$(2) \quad \mathbf{p}_t = \boldsymbol{\iota}_n m_t + \mathbf{s}_t,$$

in which the efficient price evolves as a random walk, $m_t = m_{t-1} + v_t$, with serially uncorrelated innovations v_t , and $\boldsymbol{\iota}_n$ is the n -dimensional unit vector. Market-specific microstructure effects $\mathbf{s}_t = (s_{1,t}, \dots, s_{n,t})'$ are described by a mean-zero covariance stationary process with Wold representation,

$$(3) \quad \mathbf{s}_t = \sum_{j=0}^{\infty} \boldsymbol{\Upsilon}_j \boldsymbol{\nu}_{t-j}, \quad \boldsymbol{\Upsilon}_0 = \mathbf{I}_n, \quad \sum_{j=0}^{\infty} |\boldsymbol{\Upsilon}_j| < \infty,$$

where $\boldsymbol{\nu}_t = (\nu_{1,t}, \dots, \nu_{n,t})'$, $\mathbb{E}(\boldsymbol{\nu}_t) = \mathbf{0}$, $\mathbb{E}(\boldsymbol{\nu}_t \boldsymbol{\nu}_s') = \mathbf{0} \forall s \neq t$, and $\mathbb{E}(\boldsymbol{\nu}_t \boldsymbol{\nu}_t') = \boldsymbol{\Sigma}_{\nu}$. Here, $\mathbf{I}_n$ is the identity matrix of order n . This basic microstructural framework can be linked to a reduced-form time-series representation.

Lehmann (2002) illustrates the economic content of that link by pointing out that the payoff of a zero net share trading strategy that entails buying the asset at $p_{2,t}$ and selling it immediately at $p_{1,t}$ is determined only by microstructure components, $p_{1,t} - p_{2,t} = s_{1,t} - s_{2,t}$. In econometric terms, $\boldsymbol{\beta}'_1 \mathbf{p}_t$, with $\boldsymbol{\beta}_1 = (1, -1, 0, \dots, 0)'$, is stationary, implying cointegrated prices. More generally, the $n - 1$ linearly independent rows of $\boldsymbol{\beta}' = [\boldsymbol{\iota}_{n-1} \quad -\mathbf{I}_{n-1}]$ constitute a basis for a space of cointegrating vectors.

The basic microstructural framework in equation (2) thus represents a cointegrated system of n time series (the market prices) with one common stochastic trend: the efficient price. Assuming that market price dynamics can be described

by a vector autoregression (VAR) of order q , we obtain the vector error correction model (VECM)

$$(4) \quad \Delta \mathbf{p}_t = \alpha \beta' \mathbf{p}_{t-1} + \Gamma_1 \Delta \mathbf{p}_{t-1} + \cdots + \Gamma_{q-1} \Delta \mathbf{p}_{t-q+1} + \mathbf{u}_t,$$

where α is an $(n \times n - 1)$ matrix of adjustment coefficients, and Γ_1 through Γ_{q-1} are parameter matrices (see also Engle and Granger (1987)). The serially uncorrelated, zero-mean innovations, $\mathbf{u}_t = (u_{1,t}, \dots, u_{n,t})'$, may be contemporaneously correlated with the covariance matrix $\Sigma_{\mathbf{u}}$.

As lucidly shown by Lehmann (2002), the microstructural framework in equation (2) and the VECM in equation (4) are linked by the relation

$$(5) \quad v_t = \xi' \mathbf{u}_t,$$

where ξ' is one of the identical rows of

$$(6) \quad \Xi = \beta_{\perp} \left[\alpha'_{\perp} (\mathbf{I}_n - \sum_{i=1}^{q-1} \Gamma_i) \beta_{\perp} \right]^{-1} \alpha'_{\perp},$$

with $\mathbf{a}_{\perp}$ the orthogonal complement of $\mathbf{a}$ (see also Johansen (1995)). The i th element of $\xi = (\xi_1, \dots, \xi_n)'$ thus measures the permanent impact of a price innovation $u_{i,t}$ on the efficient price m_t .

In order to divide contributions to price discovery between markets, we need to identify market-idiosyncratic price innovations. Conceptually, these are contemporaneously and serially uncorrelated standardized random variables, $\varepsilon_t = (\varepsilon_{1,t}, \dots, \varepsilon_{n,t})' \sim (\mathbf{0}, \mathbf{I}_n)$, which are linked to the correlated innovations $\mathbf{u}_t$ via the factor structure $\mathbf{u}_t = \mathbf{B} \varepsilon_t$, where $\mathbf{B}$ is a nonsingular matrix. The variance of the efficient price innovations is then given by

$$(7) \quad \text{Var}(v_t) = \text{Var}(\xi' \mathbf{B} \varepsilon_t) = \xi' \mathbf{B} \mathbf{B}' \xi.$$

The right-hand side term of equation (7) is a weighted sum of the unit variances of the idiosyncratic market innovations. This leads to the useful decomposition of the efficient price variance that constitutes Hasbrouck's (1995) information share measure. The standard procedure for identifying the n^2 elements of $\mathbf{B}$ is the Cholesky factorization of $\Sigma_{\mathbf{u}} = \mathbf{C} \mathbf{C}'$, such that $\mathbf{B}$ is equated with the lower triangular Cholesky matrix $\mathbf{C}$. The i th element of the row vector

$$(8) \quad \mathbf{IS} = \frac{[\xi' \mathbf{C}]^{(2)}}{\text{Var}(v_t)} = \frac{[\xi' \mathbf{C}]^{(2)}}{\xi' \mathbf{C} \mathbf{C}' \xi},$$

where $^{(2)}$ denotes element-wise squaring, then gives the fraction of the efficient price variance that can be attributed to market i , its *Hasbrouck information share*.

The matrix $\mathbf{C}$ is lower triangular; hence, an idiosyncratic innovation can contemporaneously affect only those markets that have a lower rank in the Cholesky ordering. Since there is no justification of an informational hierarchy of this kind, the standard solution is to permute the ordering of markets, which results in upper and lower bounds of the information shares. When these bounds diverge, as is the case in many applications (see the examples given in the Introduction), they measure contributions to price discovery very inaccurately.

B. Tail Dependence in a Microstructural Model

The identification of idiosyncratic innovations (the key to unique information shares) is a prevalent problem in various fields of economics. For that purpose, macroeconomic studies exploit theory-based restrictions on short- and/or long-run impacts of idiosyncratic shocks. However, finance theory does not suggest any constraints on the long-run effects of idiosyncratic price innovations (i.e., on $\xi' \mathbf{B}$), nor any direct constraints on $\mathbf{B}$. Hence, the indeterminacy of Hasbrouck's (1995) information share measure remained a caveat for 15 years.¹

Our solution exploits a salient feature of multivariate financial return data: tail dependence. By tail dependence we mean the phenomenon whereby the correlation between price changes in the tails of the joint distribution is different from that in the center (see also Longin and Solnik (2001)). Before we focus on the reduced-form time-series specification that will be employed to identify information shares on the basis of tail dependence, let us first show how tail-dependent price changes can emerge within the basic microstructural framework (2), in particular as a variant of Hasbrouck's (2002) stale quotes model.

Consider the case of $n = 2$ markets, in which the market 1 price incorporates the entire efficient price innovation v_t , while the market 2 price trails behind and contemporaneously accounts for only a fraction of the efficient price shock,

$$(9) \quad p_{1,t} = m_{t-1} + v_t + s_{1,t},$$

$$(10) \quad p_{2,t} = m_{t-1} + \lambda v_t + s_{2,t}, \quad 0 \leq \lambda < 1.$$

The microstructure components $s_{1,t}$ and $s_{2,t}$ are described by a covariance stationary process as in equation (3). With a reference to classic microstructural contributions like Easley and O'Hara (1992) and Easley, Kiefer, O'Hara, and Paperman (1996), let us assume that there are "no-news" periods ($n_t = 0$) during which only a modicum of new information is produced that changes the efficient price. While efficient price changes are expected to be quite small in the tranquil no-news regime, we expect them to be large in the volatile "news" regime ($n_t = 1$). Efficient price changes can thus be conceived as being drawn from a mixture of 2 mean-zero distributions (not necessarily Gaussian) with variances σ_1^2 and σ_2^2 , viz,

$$(11) \quad v_t = v_{1,t}n_t + v_{2,t}(1 - n_t),$$

$$v_{1,t} \sim (0, \sigma_1^2), \quad v_{2,t} \sim (0, \sigma_2^2), \quad \sigma_2^2 \ll \sigma_1^2 < \infty.$$

The market state may be either exogenously determined, for example, by news announcements at fixed dates, or randomly drawn such that $P(n_t = 1) = \pi$.

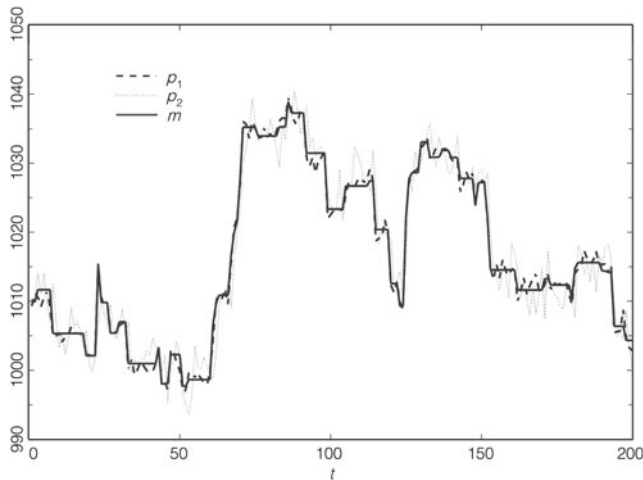
Graph A of Figure 1 depicts simulated time series of the efficient price and the market prices from a parameterized version of this model. Graph B shows the scatter plot of the corresponding market price changes. We set $\sigma_2^2 = 0$, such that the efficient price changes only in the news regime that occurs with probability $\pi = 0.3$. Efficient price changes in the news regime are drawn from a Gaussian distribution. Since market 2 contemporaneously incorporates only a fraction ($\lambda = 0.3$)

¹An attempt to resolve the problem was put forth by Lien and Shrestha (2009), who propose an alternative decomposition of Σ_u . However, the economic rationale for their decomposition is unclear.

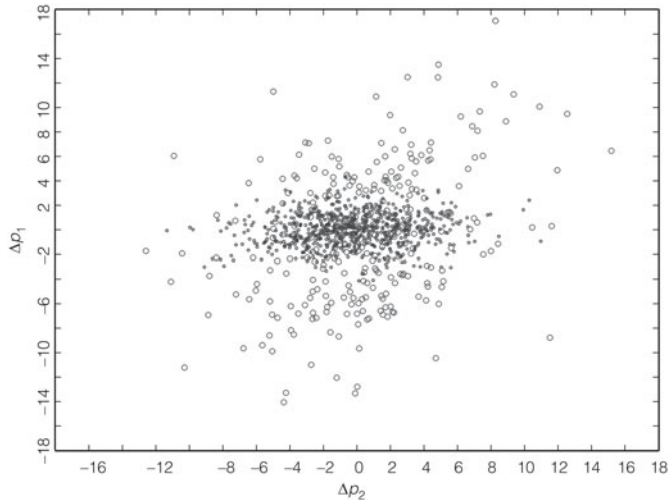
FIGURE 1
Simulated Microstructural Model Data

We simulate the microstructural model of equations (9)–(11). Efficient price innovations v_t in the “news” regime are independent draws from a mean 0 and $\sigma_v^2 = 5$ Gaussian distribution, and v_t is 0 in the “no-news” regime ($\sigma_v^2 = 0$). The news regime occurs with probability $\pi = 0.3$. The market 1 price contemporaneously incorporates 100% and the market 2 price 30% of the efficient price innovation ($\lambda = 0.3$). Market microstructure effects $s_{1,t}$ and $s_{2,t}$ are generated by 2 independent AR(2) processes, $s_{1,t} = 0.1s_{1,t-1} + 0.05s_{1,t-2} + \nu_{1,t}$, $s_{2,t} = 0.1s_{2,t-1} + 0.05s_{2,t-2} + \nu_{2,t}$, where $\nu_{1,t} \sim N(0, 1)$, $\nu_{2,t} \sim N(0, 2)$ and $\text{Cov}(\nu_{1,t}, \nu_{2,t}) = 0$ for $i \neq j$, $t \neq s$. This implies that $\text{Var}(s_{1,t}) = 1.12$ and $\text{Var}(s_{2,t}) = 2.24$. We initialize the efficient price at $m_0 = 1,000$ and simulate a time series of $T = 1,000$ market prices $p_{1,t}$, $p_{2,t}$ and efficient prices m_t . The first 200 simulated efficient prices and market prices are displayed in Graph A. Graph B shows the scatter plot of simulated price changes $\Delta p_{1,t}$, $\Delta p_{2,t}$ (all observations). The circles indicate a draw from the news regime; the dots are observations from the no-news regime.

Graph A. Price Series



Graph B. Price Changes



of the efficient price change, and since $\text{Var}(s_{1,t}) = \frac{1}{2}\text{Var}(s_{2,t})$, market 1 is both informationally more efficient and less prone to microstructure effects.

The scatter plot of simulated price changes in Graph B of Figure 1 shows little correlation in the dense center of the joint distribution. These data points tend to be associated with the no-news regime (indicated by dots), where in absolute terms large market 2 price shifts emerge as “liquidity shocks” (i.e., large draws of $|s_{2,t}|$). These noninformationally induced price shocks do not tend to be accompanied by large price changes in market 1. In the news regime (indicated by circles), however, we do observe correlated extreme price changes, as both microstructure and efficient price shocks affect the market prices: The simulated data exhibit tail dependence.

C. Information Shares through Tail Dependence

How can tail-dependent price changes be exploited to resolve the indeterminacy of the Hasbrouck (1995) information share measure? Note that the microstructural model cannot be directly employed for empirical purposes, since the efficient price is a latent variable. We therefore extend the VECM in equation (4), which maintains, via equation (5), the link to the microstructural reference framework of equation (2).

Our approach draws on Lanne and Lütkepohl’s (2010) idea of identifying structural shocks in a multiple time-series framework on the basis of nonnormal residuals, in particular by assuming a multivariate mixture distribution. Such an assumption may not be obvious or even sensible in a macroeconomic VAR involving variables such as gross domestic product (GDP) and money supply. In the present context, however, it enables us to account for tail-dependent price shocks and to provide a link to the microstructural model presented in the previous section.

Setting up the model, we maintain and extend the factor structure that connects composite and idiosyncratic market innovations. In particular, we assume $\mathbf{u}_t = \mathbf{B}\boldsymbol{\varepsilon}_t = \mathbf{W}\mathbf{e}_t$, where $\mathbf{W}$ denotes a nonsingular weighting matrix, and $\mathbf{e}_t$ is an n -dimensional vector of contemporaneously and serially uncorrelated random variables. These nonstandardized idiosyncratic innovations (which, unlike $\boldsymbol{\varepsilon}_t$, do not have unit variances) emerge as a mixture of 2 serially uncorrelated Gaussian random vectors,

$$(12) \quad \mathbf{e}_t = \begin{cases} \mathbf{e}_{1,t} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n) & \text{with probability } \gamma \\ \mathbf{e}_{2,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}) & \text{with probability } 1 - \gamma \end{cases},$$

where $\boldsymbol{\Psi}$ is a diagonal matrix with positive elements $\psi_1, \dots, \psi_n$, which we refer to as *regime variances*.

The mixture assumption and the implied regime dependence of the idiosyncratic innovation variances are consistent with the microstructural reference model of the previous section. There may be other explanations for regime-dependent model parameters (e.g., high- and low-spread market states as in Foucault’s (1999) model), which also could be used to explain tail-dependent price changes. The important insight is that the economic and statistical assumptions underlying

Hasbrouck's (1995) information shares approach remain valid for general specifications of the distribution of the efficient price innovations. We demand only the existence of the first 2 moments and the absence of serial correlation. In particular, the theoretical framework in Section II.A still applies when the idiosyncratic price innovations come from a mixture distribution as in equation (12), which can be seen by writing the covariance matrix of $\mathbf{e}_t$ as

$$(13) \quad \Sigma_{\mathbf{e}} = \gamma \mathbf{I}_n + (1 - \gamma) \Psi,$$

such that

$$(14) \quad \Sigma_{\mathbf{u}} = \mathbf{B} \mathbf{B}' = \mathbf{W} \Sigma_{\mathbf{e}} \mathbf{W}',$$

which implies that $\mathbf{B} = \mathbf{W} \Sigma_{\mathbf{e}}^{0.5}$. The vector of information shares can then be computed by replacing $\mathbf{C}$ in equation (8) with $\mathbf{W} \Sigma_{\mathbf{e}}^{0.5}$, viz,

$$(15) \quad \mathbf{IS}_T(\theta_m, \theta_v) = \frac{\left[\xi' \mathbf{W} \Sigma_{\mathbf{e}}^{0.5} \right]^{(2)}}{\xi' \mathbf{W} \Sigma_{\mathbf{e}} \mathbf{W}' \xi},$$

where the sets $\theta_v = \{\alpha, \beta, \Gamma_1, \dots, \Gamma_{q-1}\}$ and $\theta_m = \{\gamma, \Psi, \mathbf{W}\}$ collect the VECM and mixture model parameters, respectively.

The computation of information shares according to equation (15) exploits distributional properties in order to identify the weighting matrix $\mathbf{W}$. Under which conditions does the identification strategy work, and when does it fail? What is the role of tail dependence? In the following, we address these questions both intuitively and in a formal way. Before we go into technical details, let us state and illustrate the essential result in advance: The necessary condition for identifying the weighting matrix $\mathbf{W}$, and thus the $\mathbf{IS}_T$ information shares vector according to equation (15), is the tail dependence in the composite shocks $\mathbf{u}_t$, and this tail dependence emerges only if the regime variances $\psi_1, \dots, \psi_n$ are distinct. Figure 2 provides an illustration of this result in the form of scatter plots of simulated composite shocks generated by 3 bivariate parameterizations.

Visual inspection of Graphs A and B of Figure 2 scatter plots suggests that the correlation between the composite innovations in the tails of the joint distribution is more pronounced than in the center. For the Graph C data, on the other hand, the center and tail correlations do not appear to be different. The visual impression is correct. The data generating processes (DGPs) in Graphs A and B have regime variances that shift disproportionally, $\psi_1 \neq \psi_2$, while in Graph C, $\psi_1 = \psi_2$. It is straightforward to show that the correlation between the composite price innovations u_1 and u_2 is the same in both variance regimes if $\psi_1 = \psi_2$, and tail dependence occurs only when $\psi_1 \neq \psi_2$.² Figure 2 illustrates the analytical result by means of regression lines drawn through the cloud of the respective regime

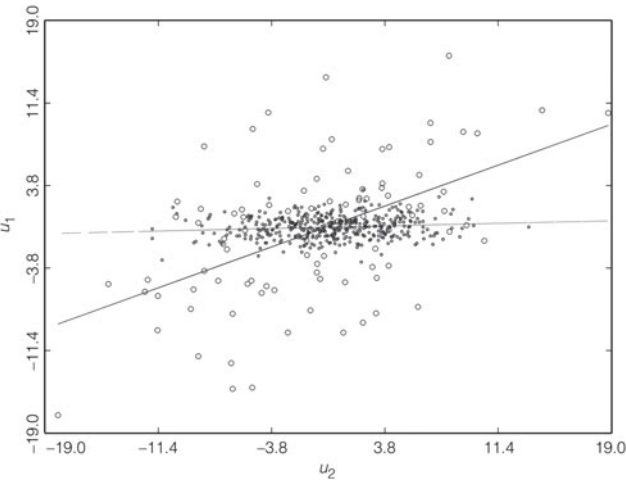
²From equation (14) it follows that the correlation between composite shocks in the 1st regime is given by

$$\rho_{u_i, u_j}^{(1)} = \frac{\sum_{k=1}^n w_{i,k} w_{j,k}}{\sqrt{\sum_{k=1}^n w_{i,k}^2} \sqrt{\sum_{k=1}^n w_{j,k}^2}},$$

FIGURE 2
Scatter Plots of Simulated Composite Price Innovations

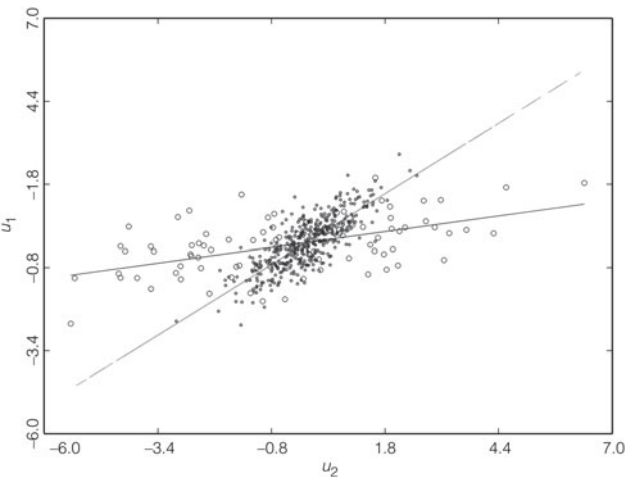
Data are generated by bivariate mixture distributions. The small dots represent observations from regime 1, the circles represent observations from regime 2. The lines result from regressions of u_1 on u_2 using data from the respective regimes. The number of draws is 500.

Graph A. Tail Dependence



$$\mathbf{w} = \begin{pmatrix} 1 & 0.05 \\ 0.5 & 4 \end{pmatrix}, \psi_1 = 40, \psi_2 = 2, \gamma = 0.8$$

Graph B. Tail Dependence

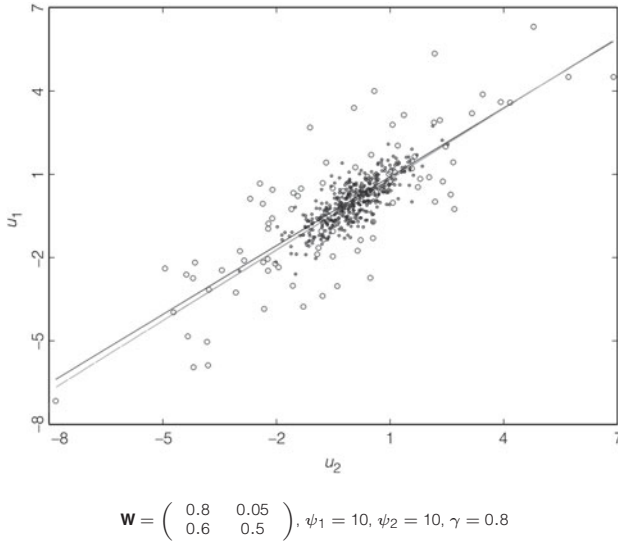


$$\mathbf{w} = \begin{pmatrix} 0.8 & 0.05 \\ 0.6 & 0.5 \end{pmatrix}, \psi_1 = 1, \psi_2 = 20, \gamma = 0.8$$

(continued on next page)

FIGURE 2 (continued)
Scatter Plots of Simulated Composite Price Innovations

Graph C. No Tail Dependence



observations. We revisit the Figure 2 data in Section II.E, where we offer an intuitive explanation of how identification on the basis of tail dependence works. However, we first need to resolve some important identification issues.

D. Information Shares Made Unique: Resolving Identification Problems

The identification of information shares using tail dependence has 2 dimensions: the determination of the *set* of information shares, and their unique *allocation* to n markets. As regards the 1st aspect, Lanne and Lütkepohl (2010) show that the columns of $\mathbf{W}$ are identified only if the matrix Ψ contains different variances on its main diagonal. We see above that this is tantamount to demanding tail-dependent price changes. The Graph C DGP in Figure 2 is thus an example in which $\mathbf{W}$ and $\mathbf{IS}_T(\theta_m, \theta_v)$ are not identified. Lanne and Lütkepohl also point out that even with distinct diagonal elements of Ψ , the columns of $\mathbf{W}$ are only identified up to a multiplication of 1 or many columns by -1 . However, this is not a serious problem, since sign shifts in the columns of $\mathbf{W}$ do not change the information shares computed according to equation (15).

and the correlation in the 2nd regime is

$$\rho_{u_i, u_j}^{(2)} = \frac{\sum_{k=1}^n w_{i,k} w_{j,k} \psi_k}{\sqrt{\sum_{k=1}^n \psi_k w_{i,k}^2} \sqrt{\sum_{k=1}^n \psi_k w_{j,k}^2}}.$$

Hence, $\rho_{u_i, u_j}^{(1)} = \rho_{u_i, u_j}^{(2)}$, when $\psi_1 = \psi_2 = \dots = \psi_n$.

More important, and fatal for the quest to deliver unique information shares, is the following fact: Distinct main diagonal elements of Ψ ensure only the identification of the *columns* of $\mathbf{W}$, but not their *order*. The consequences are severe, as this implies that it is possible only to identify the *set* of information shares, but not to *allocate* them uniquely to n markets. The allocation of information shares is the 2nd dimension of identification through tail dependence. We prove in Appendix A (Proposition 1) that there exist $n!$ observationally equivalent allocations of information shares. Observational equivalence means that these allocations imply the same joint density of the random vector $\mathbf{u}_t$ that, resulting from equation (12) and with $\mathbf{u}_t = \mathbf{W}\mathbf{e}_t$, is given by

$$(16) \quad f(\mathbf{u}_t; \theta_m) = \gamma \times (2\pi)^{-n/2} \det(\mathbf{W})^{-1} \exp \left\{ -\frac{\mathbf{u}_t'(\mathbf{W}\mathbf{W}')^{-1}\mathbf{u}_t}{2} \right\} \\ + (1 - \gamma) \times (2\pi)^{-n/2} \det(\Psi)^{-0.5} \det(\mathbf{W})^{-1} \\ \times \exp \left\{ -\frac{\mathbf{u}_t'(\mathbf{W}\Psi\mathbf{W}')^{-1}\mathbf{u}_t}{2} \right\}.$$

To be more precise, the parameterization $\theta_m^* = \{\gamma, \Psi^*, \mathbf{W}^*\}$, where $\mathbf{W}^* = \mathbf{W}\mathbf{P}$, and $\Psi^* = \mathbf{P}'\Psi\mathbf{P}$, with $\mathbf{P}$ a permutation matrix of order n , is observationally equivalent to the original parameterization $\theta_m = \{\gamma, \Psi, \mathbf{W}\}$ (i.e., $f(\mathbf{u}_t; \theta_m) = f(\mathbf{u}_t; \theta_m^*)$).³ Identification problems are aggravated, since we can find $n!$ more parameterizations $\tilde{\theta}_m = \{\tilde{\gamma}, \tilde{\Psi}, \tilde{\mathbf{W}}\}$, where $\tilde{\mathbf{W}} = \mathbf{W}\Psi^{0.5}\mathbf{P}$, $\tilde{\Psi} = \mathbf{P}'\Psi^{-1}\mathbf{P}$ and $\tilde{\gamma} = 1 - \gamma$, which are also observationally equivalent to θ_m , that is, $f(\mathbf{u}_t; \theta_m) = f(\mathbf{u}_t; \tilde{\theta}_m)$ (see also Proposition 2 in Appendix A). Altogether, there exist $2n! - 1$ parameterizations, which are observationally equivalent to θ_m . These alternative parameterizations permute the original information shares vector $\mathbf{IS}_T(\theta_m, \theta_v)$ according to

$$(17) \quad \mathbf{IS}_T(\theta_m^*, \theta_v) = \mathbf{IS}_T(\tilde{\theta}_m, \theta_v) = \mathbf{IS}_T(\theta_m, \theta_v) \times \mathbf{P},$$

such that there exist $n!$ different but observationally equivalent allocations of information shares to n markets. The formal proofs of these claims are deferred to Appendix A.

It is therefore necessary to provide economically plausible constraints that preclude the permutation of the columns of $\mathbf{W}$. Fortunately, this is possible for the *one security–multiple markets* framework. The following inequality constraints ensure a unique allocation of information shares across markets that is associated with a single parameterization:

$$(18) \quad w_{i,i} > 0, \quad \forall \quad i = 1, \dots, n,$$

$$(19) \quad w_{i,i} > |w_{j,i}|, \quad \forall \quad j \neq i,$$

$$(20) \quad \psi_i > 1, \quad \text{for an arbitrarily chosen } i,$$

³A permutation matrix $\mathbf{P}$ results from permuting the rows of an identity matrix. Including the identity matrix, there exist $n!$ distinct permutation matrices of order n . Post- (pre-) multiplication by a permutation matrix results in a matrix where the columns (rows) of a matrix are interchanged according to the permutation implied by $\mathbf{P}$. $\mathbf{W}\mathbf{P}$ thus permutes the columns of $\mathbf{W}$, while $\mathbf{P}'\Psi\mathbf{P}$ permutes the diagonal elements of Ψ accordingly.

where $w_{i,j}$ denotes the row i , column j element of $\mathbf{W}$. What do these restrictions achieve, and are they economically tenable?

By imposing the inequality constraints of equation (18), we assume that the contemporaneous effect of an idiosyncratic market i price shock, $e_{i,t}$, on the observed market i price, $p_{i,t}$, bears the same sign as the shock itself, thereby also resolving the sign indeterminacy of the columns of $\mathbf{W}$ mentioned previously. By imposing equation (19), we stipulate that the contemporaneous effect of an idiosyncratic shock originating in market i , $e_{i,t}$, on the own-market price, $p_{i,t}$, must be greater than the contemporaneous effect on all other prices. Note that the constraints in equations (18) and (19) could also be justified when the market prices do not refer to the same asset but to different securities whose valuations bear a proportional relationship. Examples are prices of American depositary receipts (ADRs), which are traded as fractions or multiples of one home market share (see also Grammig, Melvin, and Schlag (2005)), and leveraged exchange-traded funds tracking an index (see also Theissen (2012)). In these applications it is common practice to work with log prices, assuming that the log differences of the market prices are cointegrated with the log of the efficient price representing the common stochastic trend. The constraints in equation (19) then require that the *percentage* change of the own-market price in response to an idiosyncratic shock is greater than that of all other prices.

The restrictions in equations (18) and (19) associate idiosyncratic price shocks with each of the markets, fix the ordering of the columns of $\mathbf{W}$, and leave $\mathbf{P} = \mathbf{I}_n$ as the only eligible permutation matrix. The 2 remaining parameterizations, $\theta_m = \{\gamma, \Psi, \mathbf{W}\}$ and $\bar{\theta}_m = \{1 - \gamma, \Psi^{-1}, \mathbf{W}\Psi^{0.5}\}$, imply the same allocation of information shares to the n markets (see also Proposition 2 in Appendix A). Restricting one of the regime variances such that it must be greater than 1, as demanded by the inequality constraint in equation (20), leaves θ_m as the one admissible parameterization.

If the diagonal elements of Ψ are distinct, then the constraints in equations (18), (19), and (20) suffice to identify the set of information shares and assign them uniquely to the markets. Equation (15) then delivers a unique vector of *tail-dependent information shares*.

E. Identification through Tail Dependence: Intuition

Having dealt with identification issues in a formal way, we now try to provide an intuitive account of how identification based on tail dependence works. Let us see whether a visual inspection of the scatter plot of simulated composite shocks in Graph A of Figure 2 can help to reveal the structural relation

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} w_{1,1} & w_{1,2} \\ w_{2,1} & w_{2,2} \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}.$$

We take it as given that the idiosyncratic shocks e_1 and e_2 are generated from the mixture distribution in equation (12); that the identifying constraints in equations (18) and (19) hold, that is, $0 < |w_{2,1}| < w_{1,1}$, $0 < |w_{1,2}| < w_{2,2}$; and that $\psi_1 > 1$. What can we read from the scatter plot?

In the center of the empirical distribution in Graph A of Figure 2 we observe small composite innovations u_1 that are accompanied by volatile u_2 observations. These data are consistent with the interpretation that extreme shocks emerging in market 2 translate via $w_{2,2} > 0$ into a large u_2 observation of the same sign. The fact that we do not observe accompanying large u_1 observations can be explained by contemporaneous idiosyncratic innovations e_1 drawn from the low-variance regime, where $\text{Var}(e_1) = 1$, as well as by a small $w_{1,2}$ (hence, the large e_2 draw does not spill over to market 1), and a relatively small $w_{1,1}$ (compared to $w_{2,2}$). On the other hand, we do observe u_1 shocks in the tails of the distribution, accompanied by large u_2 observations bearing the same sign. These data can be explained by extreme market 1 idiosyncratic shocks occurring in the high-variance regime (i.e., when $\text{Var}(e_1) = \psi_1 \gg 1$). These shocks translate via a relatively large $w_{2,1} > 0$ into a u_2 tail observation, as well as via $w_{1,1} > 0$ into a u_1 tail observation. Interpreting the data in this way implies that $\psi_1 \gg \psi_2$.

With reference to the microstructural model developed in Section II.B, the Graph A of Figure 2 scatter plot suggests an informationally efficient market 1, where significant price shifts are associated with information processing, and a less efficient market 2 that is more prone to microstructural effects and transitory liquidity shocks. The true DGPs in the captions of Figure 2 reveal that the visual inspection indeed identifies the structural relationships correctly.

A similar visual identification works for the Graph B of Figure 2 data, too, but the scatter plot in Graph C is less revealing. Lack of tail dependence, due to identical regime variances $\psi_1 = \psi_2$, implies that the elements of the weighting matrix $\mathbf{W}$ cannot be identified.

F. Estimation and Specification Tests

1. Estimation

Maximum likelihood (ML) is the natural method with which to estimate the model parameters. Using equation (16) and $\mathbf{u}_t = A(L)\mathbf{p}_t$, where

$$(21) \quad A(L) = 1 - L - \alpha\beta'L - \Gamma_1\Delta L - \dots - \Gamma_{q-1}\Delta L^{q-1},$$

the conditional log-likelihood function reads

$$(22) \quad \mathcal{L}(\theta_m, \theta_v) = \sum_{t=1}^T \ln \left(\gamma \times (2\pi)^{-n/2} \det(\mathbf{W})^{-1} \exp \left\{ -\frac{\mathbf{u}_t'(\mathbf{W}\mathbf{W}')^{-1}\mathbf{u}_t}{2} \right\} \right. \\ \left. + (1 - \gamma) \times (2\pi)^{-n/2} \det(\Psi)^{-0.5} \det(\mathbf{W})^{-1} \right. \\ \left. \times \exp \left\{ -\frac{\mathbf{u}_t'(\mathbf{W}\Psi\mathbf{W}')^{-1}\mathbf{u}_t}{2} \right\} \right).$$

Estimation of the VECM parameters θ_v and the mixture parameters θ_m in a single step is feasible but computationally burdensome. We therefore adopt the 2-step estimation procedure outlined by Vlaar (2004). The 1st step either estimates the cointegrating vectors or uses those suggested by theory (i.e., $\beta' = [\iota_{n-1} \quad -\mathbf{I}_{n-1}]$). Equation-by-equation ordinary least squares (OLS) of the VECM in equation (4) then delivers consistent estimates of θ_v , which can be used to compute an estimate

of the vector of long-run impacts ξ according to equation (6). The 2nd estimation step maximizes the concentrated log-likelihood, which results from replacing the VECM parameters by their 1st-step estimates, that is, $\mathbf{u}_t$ in equation (22) is replaced by

$$(23) \quad \hat{\mathbf{u}}_t = (1 - L - \hat{\alpha}\hat{\beta}'L - \hat{\Gamma}_1\Delta L - \dots - \hat{\Gamma}_{q-1}\Delta L^{q-1})\mathbf{p}_t,$$

to obtain estimates of θ_m . Maximization of the concentrated log-likelihood imposes the identifying constraints in equations (18) and (19) along with $\psi_1 > 1$. Plugging the 1st-step estimates $\hat{\theta}_v$ and the 2nd-step estimates $\hat{\theta}_m$ in equation (15) delivers unique information share estimates $\widehat{\mathbf{IS}}_T(\hat{\theta}_m, \hat{\theta}_v)$. Standard errors for the estimates resulting from this 2-step procedure can be obtained using a parametric residual bootstrap along the lines of MacKinnon (2002). Details of the bootstrap procedure are provided in Appendix B.⁴

2. Specification Tests

We propose to identify information shares on the basis of tail dependence. It is therefore crucial to test whether the data support the identifying assumption. Unfortunately, the extant literature on tests for tail dependence is quite sparse. We therefore consider 2 new test ideas.

Our 1st specification test is based on the result that identification through tail dependence requires that the regime variances $\psi_1, \dots, \psi_n$ are different. One could therefore consider conducting Wald tests of the hypothesis that $\psi_i = \psi_j$ as part of the 2nd step of the estimation procedure. This is not admissible, however, because $\mathbf{W}$ is not identified under the null hypothesis, thus invalidating the 2nd estimation step. We therefore propose to construct a test statistic based on the properties of the *marginal* distributions of the VECM innovations. Consider the 2 markets case, $n = 2$. It follows from equation (16) that the marginal densities of u_1 and u_2 can be written as univariate Gaussian mixtures,

$$(24) \quad f(u_{i,t}; \gamma, w_i^*, \psi_i^*) \\ = \gamma \times (2\pi w_i^*)^{-0.5} \exp\left(\frac{u_{i,t}^2}{-2w_i^*}\right) \\ + (1 - \gamma) \times (2\pi \psi_i^* w_i^*)^{-0.5} \exp\left(\frac{u_{i,t}^2}{-2\psi_i^* w_i^*}\right), \quad i = 1, 2,$$

⁴Mixture models are sometimes difficult to pin down. One may be deluded by a “convergence” that merely reflects hitting a local optimum of the log-likelihood. We therefore recommend trying different starting values and checking to see whether the same maximum of the log-likelihood is reached. Our programs provide a safeguard in the form of a stepwise estimation procedure that estimates a sequence of simpler, nested specifications before estimating the full model. At each step, the richer parameterization will increase the likelihood value from the previous step. This facilitates estimation and prevents probing subspaces of the parameter space where local maxima may lurk. Our GAUSS programs for estimation, bootstrapping, and testing are available from the authors.

where

$$\begin{aligned} w_1^* &= w_{1,1}^2 + w_{1,2}^2, & w_2^* &= w_{2,2}^2 + w_{2,1}^2, \\ \psi_1^* &= \frac{\psi_1 w_{1,1}^2 + \psi_2 w_{1,2}^2}{w_1^*}, & \psi_2^* &= \frac{\psi_2 w_{2,2}^2 + \psi_1 w_{2,1}^2}{w_2^*}. \end{aligned}$$

Therefore, $\psi_1 = \psi_2 \iff \psi_1^* = \psi_2^*$. However, $\psi_1^* = \psi_2^*$ is a testable restriction because the parameters of the marginal densities are all identified under the null hypothesis. Note that the marginal densities in equation (24) share the same regime probability γ . This restriction can be accounted for by jointly maximizing the marginal log-likelihood functions, which is conveniently accomplished by solving

$$(25) \quad \underset{\psi_1^*, \psi_2^*, w_1^*, w_2^*, \gamma}{\operatorname{argmax}} \quad \sum_{t=1}^T \ln f(u_{1,t}; \gamma, w_1^*, \psi_1^*) + \ln f(u_{2,t}; \gamma, w_2^*, \psi_2^*).$$

The null hypothesis that $\psi_1 = \psi_2$ ($\hat{=}$ $\psi_1^* = \psi_2^*$) can then be tested using the Wald statistic

$$(26) \quad W = \frac{(\hat{\psi}_1^* - \hat{\psi}_2^*)^2}{\widehat{\operatorname{Var}}(\hat{\psi}_1^*) + 2\widehat{\operatorname{Cov}}(\hat{\psi}_1^*, \hat{\psi}_2^*) + \widehat{\operatorname{Var}}(\hat{\psi}_2^*)},$$

which is under the null hypothesis approximately $\chi^2(1)$ distributed. Extremum estimator theory applies to the estimates obtained through equation (25), such that the usual outer-product-of-gradient or Hessian-based estimates for $\operatorname{Var}(\hat{\psi}_1^*)$, $\operatorname{Var}(\hat{\psi}_2^*)$, and $\operatorname{Cov}(\hat{\psi}_1^*, \hat{\psi}_2^*)$ are readily available. In order to proceed to the estimation of the full model based on the joint density in equation (16), we should be able to reject the hypothesis that $\psi_1 = \psi_2$. If this is not possible, then the data do not provide enough evidence for tail dependence, and one should resort to information share upper and lower bounds. The test in principle also applies for $n > 2$; then the hypothesis that $\psi_i = \psi_j$ must be rejected for every combination $i \neq j$.

For our 2nd test, we adapt an idea from structural VAR analysis. If the distributional assumptions used to identify $\mathbf{W}$ are correct, then, from equations (13) and (14), we have $\boldsymbol{\Sigma}_{\mathbf{u}} = \mathbf{W}(\gamma \mathbf{I}_n + (1 - \gamma)\boldsymbol{\Psi})\mathbf{W}'$. An estimate of $\boldsymbol{\Sigma}_{\mathbf{u}}$ that is consistent irrespective of whether or not the distributional assumption is true can be computed from the 1st-step VECM residuals using $\hat{\boldsymbol{\Sigma}}_{\mathbf{u}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{u}}_t \hat{\mathbf{u}}_t'$, while the estimate $\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}^* = \hat{\mathbf{W}}(\hat{\gamma} \mathbf{I}_n + (1 - \hat{\gamma})\hat{\boldsymbol{\Psi}})\hat{\mathbf{W}}'$ will be consistent only if the assumptions identifying $\mathbf{W}$ are correct. This suggests constructing a test statistic based on the difference between $\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}$ and $\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}^*$. If the distributional assumption holds, the difference between the estimates will be small (element-wise). To obtain a scalar-valued test statistic, structural VAR specification testing suggests focusing on

$$(27) \quad \Lambda = T(\ln |\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}^*| - \ln |\hat{\boldsymbol{\Sigma}}_{\mathbf{u}}|),$$

with large values of Λ indicating misspecification. If identification was achieved by imposing short- or long-run constraints on the weighting matrix $\mathbf{B}$, then the test statistic Λ would be χ^2 under the null hypothesis that the identifying constraints are correct (Lütkepohl (2005), Ch. 9). Identification on the basis of tail

dependence, however, does not imply an analytical solution for the distribution of the test statistic in equation (27). We therefore propose bootstrapping its distribution, which is a straightforward task since the null hypothesis $\Sigma_{\mathbf{u}} = \mathbf{W}(\gamma \mathbf{I}_n + (1 - \gamma)\Psi)\mathbf{W}'$ is implicitly assumed within the parametric residual bootstrap procedure (see also Appendix B). The p -values are obtained from comparing the sample value of A with the quantiles of the bootstrap distribution.

III. Two Empirical Applications

In the following, we illustrate the benefit of our methodology with 2 empirical applications. We first revisit a research question addressed by Blanco et al. (2005), who quantify the information share of the corporate bond market and the market for credit derivatives in pricing credit risk. In Section III.B we take a look at price discovery in the recently developed spot and futures market for CO₂ emission allowances.

A. Price Discovery in the Markets for Credit Risk

Markets for credit risk are particularly interesting for the application of our methodology because movements in liquidity premia arguably explain a large proportion of the total variation in credit spreads (see also Collin-Dufresne, Goldstein, and Spencer (2001)). The microstructural reference model presented in Section II.B suggests that liquidity differences across markets may produce tail-dependent price changes, which can be exploited to compute unique information shares.

Let us briefly review instruments and structure of the market for trading credit risk. Both corporate bonds and credit derivatives, of which CDSs are the most important instruments, are traded on over-the-counter (OTC) markets. The corporate bond market determines the credit spread (p_{CS}), the difference between risky bond yields and the risk-free rate. A CDS is a contract between 2 counterparties trading credit risk. The protection buyer transfers default risk by paying an insurance fee to the protection seller willing to assume the risk. In return, the buyer receives a payoff if the underlying financial instrument defaults. The annualized fee, expressed in basis points of the notional volume, is referred to as the CDS price (p_{CDS}).

Blanco et al.'s (2005) study is an exemplary application of Hasbrouck's (1995) methodology. Since credit spread and CDS price are linked by an approximate arbitrage relation (see also Duffie (1999), Hull and White (2000), (2001)), Blanco et al. (2005) assume cointegration between p_{CDS} and p_{CS} . Accordingly, the VECM in equation (4) consists of 2 series, $p_1 = p_{CDS}$ and $p_2 = p_{CS}$, and the common stochastic trend can be interpreted as the efficient price of credit risk.

We make use of the data on CDS prices and credit spreads collected by Blanco et al. (2005). The time series of CDS prices are the midpoints of daily close-of-business indicative quotes supplied by the CDS broker CreditTrade and J.P. Morgan Securities. The CDS prices are for single-name standard International Swaps and Derivatives Association (ISDA) benchmark contracts for physical settlement, a notional volume of 10 million dollars, and 5 years' maturity, the most liquid maturity in the CDS market. Risky bond yields are from Bloomberg.

By linearly interpolating yields between 3 and 5 years and yields with more than 6½ years to maturity at the start of the sample, a 5-year yield to maturity is estimated to match the CDS maturity. Euro and dollar 5-year swap rates, respectively, are used as proxies for the risk-free rate. The resulting time series of CDS prices and credit spreads for 33 reference entities (16 U.S. and 17 European companies) run from January 2, 2001, to June 20, 2002 (383 trading days). In the following, we focus on those 26 reference entities for which the data support the existence of the hypothesized cointegrating relation (see also Table III in Blanco et al.).

Two-step estimation and specification testing are conducted as described in Section II.F.1. The 1st estimation step uses the theoretical cointegrating vector $\beta = (1, -1)'$ and $q = 2$ in the VECM in equation (4). We refrain from reporting detailed 1st-step VECM results and focus on economically more interesting indicators derived from the 1st-step estimates, in particular estimates of the upper and lower bounds of the Hasbrouck (1995) information shares (using Cholesky factorization), of long-run impact coefficients $\xi = (\xi_{\text{CDS}}, \xi_{\text{CS}})'$ and of adjustment coefficients ratios, $\kappa_{\text{CS}} = |\alpha_{\text{CS}}|/(|\alpha_{\text{CDS}}| + |\alpha_{\text{CS}}|)$. Standard errors for these estimates are obtained from the nonparametric residual bootstrap procedure proposed by Grammig et al. (2005).

Table 1 reports the 2nd-step estimates (mixture parameters) and their bootstrap standard errors along with p -values for the 2 specification tests. The test results are quite encouraging. For all reference entities, the null hypothesis that $\psi_1 = \psi_2$ is rejected at conventional significance levels. Hence, there is empirical evidence that the data support the condition that is necessary to identify *tail-dependent information shares*. Furthermore, the mixture specification survives the scrutiny of the Λ -test. The p -values for the test statistic in equation (27) are all above 10%: The unrestricted innovation covariance estimate and the estimate implied by the mixture assumption are not significantly different.

Table 2 reports the *tail-dependent information share* estimates for the CDS market along with their bootstrap standard errors and the 1st-step estimates of the Hasbrouck (1995) information shares, long-run impact coefficients and adjustment coefficients ratios.

The adjustment coefficient ratio estimate averaged across reference entities amounts to 0.84, which implies a strong adjustment of the credit spread to previous-day price differences, suggesting that the corporate bond market follows the CDS market.⁵ Furthermore, the Hasbrouck (1995) information share estimates reported in Table 2 indicate larger contributions of the CDS market to price discovery. While for some reference entities the bounds of the Hasbrouck information shares are narrow, they are quite wide for others. For instance, the lower bound of the CDS market Hasbrouck information share estimate for Ford amounts to 52.3% (standard error (s.e.) = 20.3), and the upper bound is 80.0% (s.e. = 16.9).

⁵Adjustment coefficient ratios are frequently reported in price discovery studies that deal with the *one security—two markets* framework, often with a reference to Gonzalo and Granger (1995) factor weights (see also Booth, So, and Tse (1999), Harris, McNish, and Wood (2002), and Eun and Sabherwal (2003)). However, their use as a measure of a market's contribution to price discovery has been criticized on methodological and theoretical grounds (see also Hasbrouck (1995), (2002), De Jong (2002), Lehmann (2002), and Baillie, Geoffrey, Tse, and Zabotina (2002)).

TABLE 1
Mixture Parameter Estimates and Specification Test Results

Table 1 reports estimates of the mixture parameters and specification test results. The CDS price, p_{CDS} , is the 1st series; the credit spread, p_{CS} , is the 2nd series. The values in parentheses are standard errors from the parametric residual bootstrap (see Appendix B). The columns labeled W and Δ contain the p -values (in %) for the Wald test of $\psi_1 = \psi_2$ and the bootstrapped p -values (in %) of the specification test based on equation (27), respectively.

Reference Entity	ψ_1	ψ_2	γ	$w_{1,1}$	$w_{1,2}$	$w_{2,1}$	$w_{2,2}$	W	Δ
AOL	14954.108 (1279.240)	4.851 (0.813)	0.729 (0.038)	0.086 (0.197)	-0.020 (0.029)	0.004 (0.048)	4.711 (0.207)	<0.01	36.50
Bank of Am.	363.193 (73.767)	1.423 (0.257)	0.573 (0.030)	0.203 (0.061)	0.014 (0.022)	0.032 (0.032)	3.865 (0.211)	<0.01	92.50
Bank One	32.856 (5.939)	1.365 (0.282)	0.640 (0.039)	0.737 (0.064)	-0.053 (0.065)	0.102 (0.099)	4.971 (0.262)	<0.01	82.00
Bear Stearns	119.039 (21.405)	1.900 (0.345)	0.661 (0.031)	0.587 (0.076)	0.000 (0.054)	0.014 (0.074)	5.122 (0.265)	<0.01	63.50
British Tel.	725.767 (149.080)	1.168 (0.240)	0.433 (0.035)	0.201 (0.119)	0.031 (0.037)	0.059 (0.042)	4.157 (0.277)	<0.01	85.50
Citigroup	66.632 (11.593)	0.835 (0.146)	0.649 (0.034)	0.535 (0.049)	0.010 (0.044)	0.045 (0.054)	4.652 (0.212)	<0.01	77.50
Commerzbank	488.851 (112.523)	0.663 (0.237)	0.893 (0.023)	0.151 (0.032)	-0.003 (0.011)	-0.038 (0.033)	3.229 (0.153)	0.08	65.50
Daimler	39.521 (8.166)	2.142 (0.442)	0.494 (0.043)	1.043 (0.123)	0.013 (0.139)	0.388 (0.101)	4.277 (0.334)	<0.01	83.00
Deutsche Tel.	173.237 (38.506)	3.656 (0.721)	0.369 (0.033)	0.730 (0.206)	-0.064 (0.137)	0.227 (0.073)	2.298 (0.210)	<0.01	82.50
Fiat	310.047 (64.739)	1.192 (0.296)	0.500 (0.043)	0.352 (0.149)	-0.001 (0.062)	0.074 (0.039)	2.713 (0.209)	<0.01	70.50
FleetBoston	708.814 (132.471)	1.715 (0.359)	0.551 (0.035)	0.110 (0.057)	-0.008 (0.018)	0.061 (0.044)	3.834 (0.257)	0.01	91.00
Ford	22.850 (4.188)	6.439 (1.171)	0.681 (0.037)	2.654 (0.194)	-0.199 (0.401)	0.765 (0.324)	3.655 (0.284)	0.33	91.50
GECapital	1588.611 (383.333)	1.629 (0.353)	0.866 (0.023)	0.155 (0.058)	0.002 (0.011)	0.036 (0.036)	4.734 (0.208)	<0.01	59.00
General Mot.	28.993 (5.498)	2.095 (0.433)	0.550 (0.045)	1.523 (0.151)	0.132 (0.182)	0.342 (0.137)	4.883 (0.357)	<0.01	79.00
Goldman Sachs	123.067 (21.717)	1.148 (0.215)	0.644 (0.031)	0.428 (0.055)	-0.056 (0.037)	0.015 (0.047)	4.693 (0.247)	<0.01	75.50
Iberdrola	632.948 (139.851)	5.582 (1.284)	0.885 (0.020)	0.117 (0.019)	-0.007 (0.008)	0.020 (0.040)	2.138 (0.088)	<0.01	62.50
JPMorgan	1229.124 (208.009)	4.515 (0.852)	0.481 (0.033)	0.105 (0.081)	-0.023 (0.031)	-0.011 (0.029)	2.171 (0.169)	<0.01	73.50
Lehman Bros.	82.699 (14.513)	1.067 (0.197)	0.560 (0.034)	0.612 (0.076)	-0.007 (0.060)	-0.027 (0.072)	6.210 (0.344)	<0.01	99.50
Merrill Lynch	861.935 (164.308)	0.922 (0.149)	0.662 (0.031)	0.172 (0.080)	-0.014 (0.018)	-0.008 (0.027)	4.995 (0.231)	<0.01	71.50
Metro	485.798 (103.403)	8.201 (1.821)	0.804 (0.028)	0.199 (0.046)	-0.043 (0.023)	0.022 (0.056)	2.115 (0.112)	<0.01	67.50
Morgan Stanley	345.376 (65.204)	2.220 (0.402)	0.679 (0.028)	0.293 (0.068)	-0.021 (0.027)	0.025 (0.040)	4.029 (0.210)	<0.01	78.00
Siemens	1393.361 (250.220)	4.135 (0.817)	0.695 (0.036)	0.098 (0.075)	-0.010 (0.021)	0.022 (0.039)	2.223 (0.130)	<0.01	79.00
Telefónica	567.298 (117.300)	1.463 (0.248)	0.623 (0.030)	0.275 (0.105)	0.004 (0.031)	0.044 (0.023)	2.291 (0.118)	<0.01	82.00
Volvo	225.962 (40.362)	2.786 (0.481)	0.689 (0.029)	0.460 (0.076)	0.075 (0.042)	0.052 (0.030)	2.124 (0.109)	<0.01	79.00
Walmart	408.636 (84.480)	2.157 (0.526)	0.882 (0.020)	0.138 (0.018)	-0.007 (0.009)	0.054 (0.055)	4.087 (0.174)	<0.01	55.50
Wells Fargo	82.850 (14.654)	1.418 (0.305)	0.749 (0.030)	0.496 (0.044)	0.000 (0.038)	-0.022 (0.073)	4.529 (0.223)	<0.01	66.00

TABLE 2
Price Discovery in the Markets for Credit Risk

Table 2 reports the adjustment coefficient ratio,

$$\kappa_{CS} = \frac{|\alpha_{CS}|}{|\alpha_{CS}| + |\alpha_{CDS}|},$$

long-run impact coefficients, ξ_{CDS} and ξ_{CS} , Hasbrouck (1995) information shares for the CDS market (lower bound, upper bound, midpoint), and *tail-dependent information shares* for the CDS market ($IS_T(CDS)$). The information share estimates are expressed in %. The values in parentheses are bootstrap standard errors. Reference entities are sorted in descending order of the width of the information share bounds.

Reference Entity	κ_{CS}	ξ_{CDS}	ξ_{CS}	Hasbrouck IS(CDS)			$IS_T(CDS)$
				Low	Up	Mid	
Ford	0.58 (0.21)	0.31 (0.09)	0.22 (0.08)	52.3 (20.3)	80.0 (16.9)	66.2 (18.3)	83.4 (16.8)
Daimler	0.81 (0.23)	0.48 (0.12)	0.12 (0.11)	71.3 (21.8)	94.1 (14.2)	82.7 (17.6)	93.9 (13.8)
Telefónica	0.59 (0.25)	0.29 (0.12)	0.20 (0.10)	65.9 (23.3)	87.0 (20.0)	76.4 (21.4)	86.9 (18.0)
Fiat	0.79 (0.20)	0.45 (0.16)	0.11 (0.14)	79.4 (20.2)	97.6 (12.5)	88.5 (15.8)	97.6 (9.1)
General Mot.	0.75 (0.23)	0.40 (0.07)	0.14 (0.06)	72.1 (16.0)	90.1 (11.0)	81.1 (13.1)	88.3 (11.0)
Volvo	0.52 (0.22)	0.26 (0.08)	0.24 (0.07)	59.0 (20.4)	76.5 (17.8)	67.8 (18.8)	74.5 (18.0)
British Tel.	0.87 (0.14)	0.50 (0.15)	0.07 (0.12)	82.9 (20.9)	97.8 (13.5)	90.3 (16.7)	97.5 (13.1)
FleetBoston	0.93 (0.05)	0.48 (0.07)	0.04 (0.05)	84.2 (19.4)	97.1 (10.7)	90.7 (14.3)	97.3 (9.6)
Commerzbank	0.82 (0.24)	0.42 (0.07)	0.09 (0.04)	69.0 (20.9)	76.9 (22.7)	72.9 (21.6)	69.2 (23.0)
Walmart	0.85 (0.03)	0.41 (0.08)	0.07 (0.03)	58.1 (21.2)	65.8 (20.5)	62.0 (20.7)	66.5 (19.6)
Siemens	0.88 (0.23)	0.45 (0.10)	0.06 (0.08)	88.7 (18.0)	95.8 (14.0)	92.2 (15.8)	96.1 (13.3)
Deutsche Tel.	0.72 (0.16)	0.83 (0.99)	−0.33 (0.89)	89.6 (23.7)	95.2 (9.5)	92.4 (14.5)	94.6 (8.9)
Iberdrola	0.77 (0.23)	0.40 (0.07)	0.12 (0.05)	58.9 (22.2)	64.5 (21.8)	61.7 (21.9)	65.3 (20.5)
Citigroup	0.72 (0.15)	0.33 (0.04)	0.13 (0.03)	65.8 (14.9)	70.7 (13.7)	68.3 (14.0)	70.4 (13.6)
Bank One	0.69 (0.17)	0.30 (0.03)	0.14 (0.03)	51.5 (16.7)	56.0 (16.2)	53.8 (16.2)	58.2 (14.8)
Bank of Am.	0.93 (0.06)	0.43 (0.09)	0.03 (0.09)	95.5 (17.0)	98.7 (13.5)	97.1 (15.0)	98.5 (11.5)
Morgan St.	0.83 (0.17)	0.42 (0.06)	0.09 (0.06)	88.4 (16.1)	91.3 (14.2)	89.8 (15.0)	91.7 (12.8)
Wells Fargo	0.75 (0.11)	0.33 (0.04)	0.11 (0.04)	67.1 (18.4)	68.1 (17.2)	68.1 (17.2)	67.1 (15.6)
Lehman Bros.	0.80 (0.11)	0.39 (0.04)	0.10 (0.03)	84.0 (10.9)	86.0 (11.1)	85.0 (10.9)	84.2 (10.1)
GECapital	0.98 (0.16)	0.53 (0.08)	0.01 (0.05)	97.8 (10.1)	99.8 (7.5)	98.8 (8.6)	99.8 (7.9)
Metro	0.88 (0.19)	0.43 (0.07)	0.06 (0.06)	94.0 (15.0)	95.4 (13.8)	94.7 (14.2)	96.8 (12.4)
Bear Stearns	0.78 (0.16)	0.36 (0.04)	0.10 (0.04)	82.5 (14.1)	83.7 (13.1)	83.1 (13.4)	83.7 (12.1)
Merrill Lynch	0.90 (0.26)	0.43 (0.05)	0.05 (0.04)	96.5 (8.7)	97.7 (9.3)	97.1 (8.9)	96.7 (9.7)
JPMorgan	0.94 (0.12)	0.47 (0.09)	0.03 (0.08)	99.2 (11.6)	100.0 (13.7)	99.6 (12.4)	99.4 (10.5)

(continued on next page)

TABLE 2 (continued)
Price Discovery in the Markets for Credit Risk

Reference Entity	κ_{CS}	ξ_{CDS}	ξ_{CS}	Hasbrouck IS(CDS)			IS τ (CDS)
				Low	Up	Mid	
AOL	0.97 (0.23)	0.48 (0.08)	0.01 (0.07)	99.6 (6.4)	99.9 (5.2)	99.7 (5.7)	99.9 (6.9)
Goldman Sachs	0.80 (0.21)	0.37 (0.04)	0.10 (0.04)	84.1 (12.9)	84.1 (12.6)	84.1 (12.5)	85.6 (11.5)
Mean	0.81	0.42	0.08	78.4	86.6	82.5	86.3
Std. dev.	0.11	0.10	0.10	15.2	12.9	13.5	13.0

The last column in Table 2 reports the estimates of the CDS market information shares identified on the basis of tail dependence. For the reference entity Ford, the *tail-dependent information share* estimate amounts to 83.4% (s.e. = 16.8), a value close to the upper bound of the Hasbrouck (1995) information share. The pronounced informational leadership of the CDS market holds as a general result. For those reference entities with wide bounds, the *tail-dependent information shares* of the CDS market tend to be close to the Hasbrouck information share upper bounds.

To gain an understanding of the result, it is instructive to take a look at the scatter plots of the VECM residuals depicted in Figure 3. The residual scatter plots resemble those of the price change data from the microstructural model simulation in Figure 1 and the scatter plot of the simulated price innovations in Graph A of Figure 2. The 4 graphs show horizontally flattened dense centers of the bivariate distributions, which imply that tail observations for the credit spread residuals do not tend to be accompanied by extreme CDS residuals. However, when the CDS residual is large and positive (negative), the credit spread residual tends to be large and positive (negative), too. This pattern complies with the interpretation of a corporate bond market in which extreme but transitory price shocks may occur due to a lack of liquidity. Price innovations in the more liquid CDS market, on the other hand, tend to convey information with respect to the price of credit risk that spills over contemporaneously to the corporate bond market. Yan and Zivot (2010) argue that liquidity attracts trading volume and promotes a market's leadership in price discovery. Our findings suggest that this conclusion also holds for markets trading credit risk.

The estimates of the weighting matrix $\mathbf{W}$ reported in Table 1 correspond to this account. The estimate of $w_{2,1}$, the weight with which an idiosyncratic CDS innovation contemporaneously affects the credit spread, tends to be larger than the estimate of $w_{1,2}$, the weight with which an idiosyncratic credit spread innovation contemporaneously affects the CDS price. The estimate of $w_{1,2}$ is in most cases not significantly different from 0. The relative illiquidity of the corporate bond market thus helps to identify contemporaneous effects and facilitates the estimation of *tail-dependent information shares*.

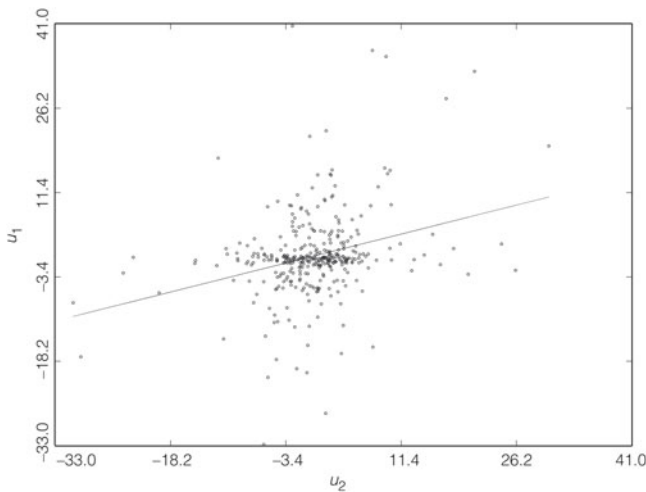
For some of the reference entities, the bounds of the Hasbrouck (1995) information shares are narrow because the contemporaneous correlations between credit spread and CDS price residuals are small. In these instances, the estimates of the off-diagonal elements of $\mathbf{W}$ are small and not significantly different from 0,

and the *tail-dependent information shares* are close to the Hasbrouck information share midpoints. We take it as a sign of robustness that the standard identification method and identification through tail dependence deliver very similar results when no ambiguity prevails in terms of wide bounds. Furthermore, the estimation accuracy in terms of standard errors is comparable. Hence, the increase in precision offered by our methodology is unambiguous.

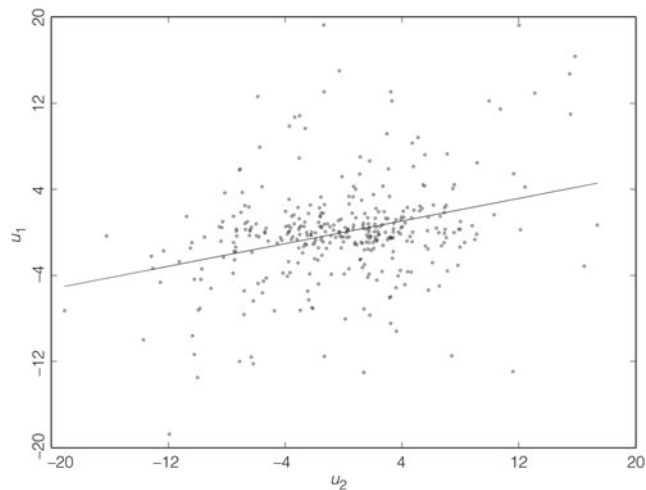
FIGURE 3
Scatter Plots of VECM Residuals

Graphs A–D of Figure 3 show scatter plots of residuals from the 1st-step VECM estimation for 4 reference entities. Here, u_1 are CDS residuals, u_2 are credit spread residuals. The lines result from a regression of u_1 on u_2 .

Graph A. Ford



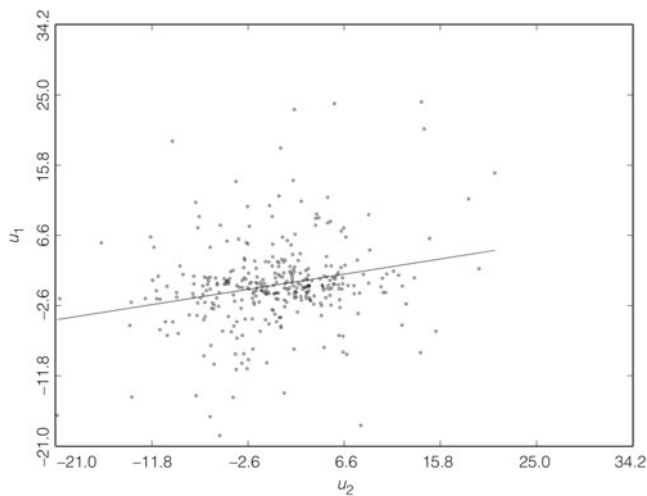
Graph B. Daimler



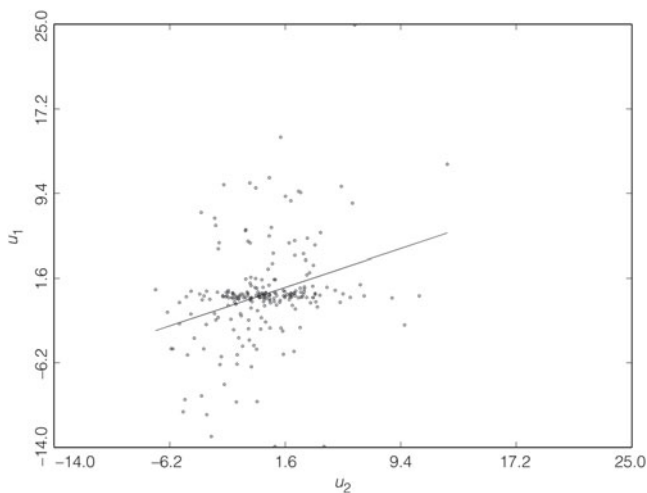
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FIGURE 3 (continued)
Scatter Plots of VECM Residuals

Graph C. General Motors



Graph D. Fiat



B. Price Discovery in the CO₂ Emission Allowances Markets

Our 2nd application revisits a research question addressed by Uhrig-Homburg and Wagner (2009), who examine price discovery in the futures and spot markets of European greenhouse gas emission allowances (EUAs). EUAs have been tradable since 2005 within the European Union emissions trading scheme. The analysis of this newly developed market, which finished its 1st trial period in 2007, is of particular interest for investors and regulatory authorities. The Uhrig-Homburg and Wagner study is another exemplary application of the Hasbrouck (1995) methodology. While the authors report information share midpoints indicating

that informational leadership lies with the futures market, again a large spread between the lower and upper information share bounds blurs the result. Can *tail-dependent information shares* help sharpen their conclusions?

To answer this question, we use the database assembled by Uhrig-Homburg and Wagner (2009), which comprises daily settlement prices of the futures and spot markets from June 24, 2005, to November 15, 2006. EUA spot prices are provided by the Powernext exchange, and futures prices are released by the European Climate Exchange. EUA trading on these exchanges is organized in continuous trading sessions and supported by market makers.

We follow Uhrig-Homburg and Wagner (2009) and compute theoretical futures prices of EUAs using a cost-of-carry model that neglects convenience yields,

(28)
$$TF_t = e^{r(\tau-t)}S_t,$$

where S_t denotes the spot price and TF_t the theoretical futures price of a contract with delivery in τ , and r gives the corresponding risk-free rate. The risk-free rate is proxied by the Euribor midquotes for maturities up to 1 year and the EuroSwap midquotes for maturities above 1 year. Among the possible futures contracts, Uhrig-Homburg and Wagner select the one maturing in December, the most frequently traded contract.

Observed futures prices F_t should deviate from TF_t only due to transitory microstructure effects. With reference to the basic microstructural framework in equation (2), we associate $p_{1,t}$ with F_t and $p_{2,t}$ with TF_t . Two-step estimation and specification testing is carried out as outlined in Section II.F.1. For the 1st step we use the theoretical cointegrating vector $\beta = (1, -1)'$ and $q = 2$ in equation (4).

Table 3 reports estimation and test results. The futures information share midpoint estimate amounts to 65%, indicating informational leadership on the part of the futures market. However, the 64-percentage-point spread between the information share lower and upper bounds blurs the conclusions.

TABLE 3
Price Discovery in the CO₂ Emission Allowances Markets

Table 3 reports concentrated ML estimates of the mixture parameters. The futures price is the 1st series, and the theoretical futures price derived from spot prices according to equation (28) is the 2nd series. The columns labeled W and Λ contain the p -value (in %) for the Wald test of $\psi_1 = \psi_2$ and the bootstrapped p -value (in %) of the specification test based on equation (27), respectively. The table also reports the Hasbrouck (1995) information shares (IS(F)) lower and upper bounds and their midpoint, as well as the *tail-dependent information share* of the futures market, $IS_T(F)$. The information share estimates are expressed in %. The values in parentheses are bootstrap standard errors.

ψ_1	ψ_2	γ	$w_{1,1}$	$w_{1,2}$	$w_{2,1}$	$w_{2,2}$	W	Λ	Hasbrouck IS(F)			
									Low	Up	Mid	IS τ (F)
140.036 (56.973)	22.647 (9.837)	0.950 (0.015)	0.212 (0.096)	0.412 (0.243)	0.086 (0.073)	0.441 (0.039)	19.86	88.26	32.60 (13.24)	96.73 (3.43)	64.67 (5.36)	68.08 (18.07)

Tail-dependent information shares may help to sharpen the result, but we first have to see whether the data support the identifying assumption. In comparison to the 1st application, however, the specification test results are less supportive for identification on the basis of tail dependence. With a p -value of 20%, the difference between the estimated regime variances is not significant on conventional levels. This suggests that either tail dependence is too weak to obtain a significant

result or that tail dependence is not present in the first place. In the latter case, the matrix $\mathbf{W}$ and consequently *tail-dependent information shares* would not be identified. The model specification test based on the A -statistic in equation (27) yields a more favorable result in that the difference between the model-free and the model-implied innovation covariance matrix estimates is not significant.

Since the test results are ambiguous, and nonidentification may be a looming issue, the 2nd-step estimates and, derived from them, the *tail-dependent information share* estimates should be interpreted with caution. Table 3 indicates that the futures market *tail-dependent information share* estimate amounts to 68%, a number close to the Hasbrouck (1995) information share midpoint of 65%. However, the standard error for the *tail-dependent information share* estimate is considerably larger than the standard error of the Hasbrouck information share midpoint. Note also that the difference between the estimated regime variances (from the full model) is considerable, but the large standard errors explain why the Wald statistic cannot reject the hypothesis that $\psi_1 = \psi_2$. Overall, the 2nd-step results indicate that tail dependence is not strong enough to obtain precise results.

In instances like this, we recommend that some prudence be exercised when identifying information shares on the basis of tail dependence. In the present application, the 2nd-step estimation results are plausible, and the large standard errors do not paint a euphemistic picture of precision. However, one should keep in mind that if there is no tail dependence in the first place, then the object of interest, the information share vector in equation (15), will not be identified.

Our 2nd application shows that identification on the basis of tail dependence is no panacea, and that there may be situations in which we have to put up with wide information share bounds. The data may just be insufficient to yield a more precise answer. We recommend that the specification test results, especially those of the Wald test for identical regime variances, should be taken very seriously before estimating *tail-dependent information shares*.

IV. Conclusion

“Where does price discovery take place?” is a key question in empirical finance. It is frequently raised in the context of studies on the competition for order flow between traditional and alternative trading platforms, national and international exchanges, and parallel markets for traditional and innovative financial instruments.

Hasbrouck’s (1995) methodology is the standard approach to addressing this research question empirically. He proposes to estimate the information share for each of the parallel markets on which financial instruments linked by the law of one price are traded. Information shares result from decomposing the variance of the innovations to the market prices’ common stochastic trend, the efficient price of the security.

The competitive edge of Hasbrouck’s (1995) information shares over alternative methodologies used to measure contributions to price discovery is widely accepted (see also Lehmann (2002)). However, most applications suffer from a lack of identification, since the contemporaneous dependence structure of market price innovations cannot be disentangled without further restrictions. As a

solution, Hasbrouck (1995) suggests a Cholesky decomposition of the covariance matrix of the market price innovations, thereby assuming a hierarchical ordering of markets that is hardly ever justifiable. In empirical work, researchers often resort to permuting the ordering of the markets, which yields upper and lower bounds of information shares rather than a unique measure. These bounds can be so wide that it is impossible to determine even the leading market.

The present paper resolves the problem of indeterminate information shares by exploiting the informational content of distributional properties of financial prices. We show that different correlations of contemporaneous price innovations in the tails and in the center of the distributions deliver the information necessary to determine unique information shares. Tail dependence of this kind can be caused by the design of the trading process, which may induce market-specific liquidity effects. Since in most applications of the Hasbrouck (1995) methodology the market structures are clearly different—which is why alternative trading platforms emerge in the first place—our methodology presents an appealing solution. With regard to the pricing of credit risk, it is the higher liquidity of the CDS market compared to the corporate bond market that sharpens the finding that informational leadership lay with the credit derivatives market during the precrisis period.

We have applied the new methodology to data sampled on a low (daily) frequency at which the contemporaneous correlation of market price innovations, the source of the indeterminacy of Hasbrouck's (1995) information share measure, is strong. Indeed, many interesting securities trade on OTC dealer markets that may generate a public quote only once a day. If shortening the sampling interval is possible, the indeterminacy problem will usually be mitigated, because then the contemporaneous correlation tends to be smaller. However, the advantages of ultra-high-frequency data have to be balanced against their drawbacks: Veritable microstructure noise, like inaccurate timing of quotes, may bias the results. Overall, the methodology presented in this paper seems to be best suited for low- to medium-frequency data where going to a fine interval of observation is either impossible or not advisable. When quality ultra-high-frequency data are available, intensity-based models, which account for the irregular spacing of the data in time (see also Hautsch (2004)), could be employed to filter price-relevant events on parallel markets and to construct alternative measures of price discovery. Kehrle and Peter (2013) take a 1st step in that direction, but a coherent link to the efficient price process, comparable to Hasbrouck's (1995) framework, has yet to be established. An interesting extension of the idea to identify information shares through tail dependence, which seems useful when a high sampling frequency still implies wide information bounds, would be to allow for a dependence of regimes over time. Extending the Markov-switching framework proposed by Lanne, Lütkepohl, and Maciejowska (2010) could be considered for that purpose.

The relation between market liquidity and contributions to price discovery has recently been emphasized by Yan and Zivot (2010). The methodology that we propose in this paper systematically exploits the informational content of those market design effects and delivers a unique measure of a market's information share. Researchers concerned with quantifying contributions to price discovery have a new tool to sharpen their conclusions.

Appendix A. Identification: Propositions and Proofs

Proposition 1. Let $\theta_m = \{\gamma, \Psi, \mathbf{W}\}$ denote the set of mixture parameters that yields the density of $f(\mathbf{u}_t; \theta_m)$ given in equation (16), and $\theta_v = \{\alpha, \beta, \Gamma_1, \dots, \Gamma_q\}$ a set of VECM parameters. Suppose the main diagonal elements of $\mathbf{W}$ are all greater than 0, and that the elements of the diagonal matrix Ψ are distinct. Furthermore, let $\mathbf{IS}_T(\theta_m, \theta_v)$ denote the vector of information shares given by equation (15). Then, holding the mixture probability γ fixed, there exist $n! - 1$ further sets of mixture parameters $\theta_m^* = \{\gamma, \Psi^*, \mathbf{W}^*\}$ given by $n! - 1$ distinct permutations of the columns in $\mathbf{W}$ and the corresponding elements in Ψ ,

$$(A-1) \quad \Psi^* = \mathbf{P}'\Psi\mathbf{P},$$

$$(A-2) \quad \mathbf{W}^* = \mathbf{W}\mathbf{P},$$

where $\mathbf{P}$ is a permutation matrix of order n . The parameterizations θ_m^* are observationally equivalent to θ_m in that

$$(A-3) \quad f(\mathbf{u}_t; \theta_m) = f(\mathbf{u}_t; \theta_m^*),$$

and they permute the original vector of information shares according to

$$(A-4) \quad \mathbf{IS}_T(\theta_m^*, \theta_v) = \mathbf{IS}_T(\theta_m, \theta_v) \times \mathbf{P}.$$

Proof. To prove the 1st part of Proposition 1, note that the observational equivalence of 2 mixture parameterizations $\theta_m = \{\gamma, \Psi, \mathbf{W}\}$ and $\theta_m^* = \{\gamma, \Psi^*, \mathbf{W}^*\}$ entails that the 2 covariance matrices of the composite shocks $\mathbf{u}_t$, $\gamma\mathbf{W}\mathbf{W}' + (1 - \gamma)\mathbf{W}\Psi\mathbf{W}'$ and $\gamma\mathbf{W}^*\mathbf{W}^{*'} + (1 - \gamma)\mathbf{W}^*\Psi^*\mathbf{W}^{*'}$ are identical. Hence, let $\mathbf{Q}$ be a matrix, such that $\mathbf{W}^* = \mathbf{W}\mathbf{Q}$, and Ψ^* be a diagonal matrix with distinct positive elements. Then $f(\mathbf{u}_t; \theta_m) = f(\mathbf{u}_t; \theta_m^*)$ implies that

$$(A-5) \quad \mathbf{W}[\gamma\mathbf{I}_n + (1 - \gamma)\Psi]\mathbf{W}' = \mathbf{W}\mathbf{Q}[\gamma\mathbf{I}_n + (1 - \gamma)\Psi^*]\mathbf{Q}'\mathbf{W}'.$$

Multiplying equation (A-5) from the left by $\mathbf{W}^{-1}$ and from the right by its transpose and rearranging terms yields

$$(A-6) \quad \gamma(\mathbf{I}_n - \mathbf{Q}\mathbf{Q}') = (1 - \gamma)(\mathbf{Q}\Psi^*\mathbf{Q}' - \Psi).$$

This holds for $0 < \gamma < 1$ only if both sides of equation (A-6) are 0, which implies that

$$(A-7) \quad \Psi = \mathbf{Q}\Psi^*\mathbf{Q}'$$

and

$$(A-8) \quad \mathbf{Q}\mathbf{Q}' = \mathbf{I}_n.$$

It follows from equation (A-8) that $\mathbf{Q}$ must be orthogonal (i.e., $\mathbf{Q}' = \mathbf{Q}^{-1}$). Hence, equation (A-7) can be regarded as a spectral decomposition of Ψ , where Ψ^* contains the eigenvalues of Ψ on its diagonal, and the columns of $\mathbf{Q}$ are the corresponding eigenvectors. As all elements of Ψ are assumed to be distinct, the columns of $\mathbf{Q}$ are linearly independent, unit-length vectors. Consequently, all possible solutions for $\mathbf{Q}$ are given by $\mathbf{Q} = \mathbf{P}\mathbf{S}$, where $\mathbf{P}$ is an n -dimensional permutation matrix and $\mathbf{S}$ an n -dimensional diagonal matrix, whose diagonal elements are either 1 or -1 . Therefore, $\mathbf{W}^* = \mathbf{W}\mathbf{P}\mathbf{S}$, which implies that the columns of $\mathbf{W}$ are identified up to multiplication by -1 . However, as the main diagonal elements of $\mathbf{W}$ are restricted such that they must be greater than 0, only $\mathbf{S} = \mathbf{I}_n$ is eligible, which yields equation (A-2). This implies that there exist $n!$ permutations of the columns in $\mathbf{W}$, of which $n! - 1$ yield a matrix $\mathbf{W}^*$ that is distinct from $\mathbf{W}$. The only permutation

matrix that leaves the ordering of the columns in $\mathbf{W}$ unchanged is $\mathbf{P} = \mathbf{I}_n$. From equation (A-7) it follows that

$$(A-9) \quad \boldsymbol{\Psi} = \mathbf{P}\mathbf{S}\boldsymbol{\Psi}^*\mathbf{S}'\mathbf{P}' = \mathbf{P}\boldsymbol{\Psi}^*\mathbf{P}'.$$

Solving for $\boldsymbol{\Psi}^*$ yields equation (A-1). Here, $\boldsymbol{\Psi}^* = \mathbf{P}'\boldsymbol{\Psi}\mathbf{P}$ is a diagonal matrix, which results from a permutation of the diagonal elements of $\boldsymbol{\Psi}$. This proves the 1st part of Proposition 1.

To prove equation (A-4), start from equation (15), which, written in detail, reads

$$(A-10) \quad \mathbf{IS}_T(\theta_m, \theta_v) = \frac{[\boldsymbol{\xi}'\mathbf{W}(\gamma\mathbf{I}_n + (1 - \gamma)\boldsymbol{\Psi})^{0.5}]^{(2)}}{\boldsymbol{\xi}'\mathbf{W}(\gamma\mathbf{I}_n + (1 - \gamma)\boldsymbol{\Psi})\mathbf{W}'\boldsymbol{\xi}}.$$

Since equation (A-5) holds, θ_m and θ_m^* imply the same covariance matrix of $\mathbf{u}_t$, and the denominator in equation (A-10) is not affected by the permutation of elements in $\mathbf{W}$ and $\boldsymbol{\Psi}$ according to equations (A-1) and (A-2). Therefore, $\mathbf{IS}_T(\theta_m^*, \theta_v)$ and $\mathbf{IS}_T(\theta_m, \theta_v)$ can differ from one another only in their numerators, which relate to each other by

$$\begin{aligned} [\boldsymbol{\xi}'\mathbf{W}^*(\gamma\mathbf{I}_n + (1 - \gamma)\boldsymbol{\Psi}^*)^{0.5}]^{(2)} &= [\boldsymbol{\xi}'\mathbf{W}\mathbf{P}(\gamma\mathbf{I}_n + (1 - \gamma)\mathbf{P}'\boldsymbol{\Psi}\mathbf{P})^{0.5}]^{(2)} \\ &= [\boldsymbol{\xi}'\mathbf{W}\mathbf{P}\mathbf{P}'\boldsymbol{\Sigma}_e^{0.5}\mathbf{P}]^{(2)} \\ &= [\boldsymbol{\xi}'\mathbf{W}'\boldsymbol{\Sigma}_e^{0.5}]^{(2)}\mathbf{P}. \end{aligned}$$

Thus, $\mathbf{IS}_T(\theta_m^*, \theta_v) = \mathbf{IS}_T(\theta_m, \theta_v)\mathbf{P}$, such that $\mathbf{IS}_T(\theta_m^*, \theta_v) \neq \mathbf{IS}_T(\theta_m, \theta_v) \forall \mathbf{P} \neq \mathbf{I}_n$. This leaves $n! - 1$ distinct permutation matrices $\mathbf{P}$ associated with $n! - 1$ different sets of mixture parameters, which are observationally equivalent but imply different information share vectors. Thereby, the proposition is proven. $\square$

Proposition 2. Let $\theta_m = \{\gamma, \boldsymbol{\Psi}, \mathbf{W}\}$ denote the set of mixture parameters that yields the density of $f(\mathbf{u}_t; \theta_m)$ given in equation (16), and $\theta_v = \{\boldsymbol{\alpha}, \boldsymbol{\beta}, \mathbf{I}_1, \dots, \mathbf{I}_q\}$ a set of VECM parameters. Suppose that the elements of the diagonal matrix $\boldsymbol{\Psi}$ are distinct. Furthermore, let $\mathbf{IS}_T(\theta_m, \theta_v)$ denote the vector of information shares given by equation (15). If it holds for the elements of $\mathbf{W}$ that

$$(A-11) \quad \begin{aligned} w_{i,i} &> 0, \quad \forall \quad i, \\ w_{i,i} &> |w_{j,i}|, \quad \forall \quad j \neq i, \end{aligned}$$

then there exists only one set of mixture parameters $\bar{\theta} = \{\bar{\gamma}, \bar{\boldsymbol{\Psi}}, \bar{\mathbf{W}}\}$, given by

$$(A-12) \quad \begin{aligned} \bar{\gamma} &= 1 - \gamma, \\ \bar{\boldsymbol{\Psi}} &= \boldsymbol{\Psi}^{-1}, \\ \bar{\mathbf{W}} &= \mathbf{W}\boldsymbol{\Psi}^{0.5}, \end{aligned}$$

that is observationally equivalent to θ_m in that $f(\mathbf{u}_t; \theta_m) = f(\mathbf{u}_t; \bar{\theta}_m)$. Furthermore, the parameterization $\bar{\theta}_m$ implies that

$$(A-13) \quad \mathbf{IS}_T(\bar{\theta}_m, \theta_v) = \mathbf{IS}_T(\theta_m, \theta_v).$$

Proof. Let $\mathbf{Q}$ be a matrix, such that $\bar{\mathbf{W}} = \mathbf{W}\mathbf{Q}$; then, it must hold that

$$(A-14) \quad \mathbf{W}[\gamma\mathbf{I}_n + (1 - \gamma)\boldsymbol{\Psi}]\mathbf{W}' = \mathbf{W}\mathbf{Q}[(1 - \gamma)\mathbf{I}_n + \gamma\bar{\boldsymbol{\Psi}}]\mathbf{Q}'\mathbf{W}'.$$

Multiplying equation (A-14) from the left by $\mathbf{W}^{-1}$ and from the right by its transpose and rearranging terms yields

$$(A-15) \quad \gamma(\mathbf{I}_n - \mathbf{Q}\bar{\boldsymbol{\Psi}}\mathbf{Q}') = (1 - \gamma)(\mathbf{Q}\mathbf{Q}' - \boldsymbol{\Psi}).$$

This holds only if both sides of equation (A-15) are 0, which implies that $\mathbf{Q}\bar{\boldsymbol{\Psi}}\mathbf{Q}' = \mathbf{I}_n$ and $\mathbf{Q}\mathbf{Q}' = \boldsymbol{\Psi}$. The latter equation gives $\mathbf{Q} = \boldsymbol{\Psi}^{0.5}$, and using this result for the first yields

$\bar{\Psi} = \Psi^{-1}$ and $\bar{\mathbf{W}} = \mathbf{W}\Psi^{0.5}$, which shows that equation (A-12) holds. The restrictions in equation (A-11) rule out permuting the columns of $\bar{\mathbf{W}}$ and the diagonal elements of $\bar{\Psi}$. Without these restrictions, $\bar{\mathbf{W}} = \bar{\mathbf{W}}\mathbf{P}$ and $\bar{\Psi} = \mathbf{P}\bar{\Psi}\mathbf{P}'$ with $\mathbf{P} \neq \mathbf{I}_n$ would yield $n! - 1$ observationally equivalent parameterizations. Thereby, the 1st part of Proposition 2 is proven.

Since equation (A-14) holds, the vector $\mathbf{IS}_T(\bar{\theta}_m)$ can differ from $\mathbf{IS}_T(\theta_m, \theta_v)$ only in terms of the vectors in the numerators, but

$$\begin{aligned}
 \text{(A-16)} \quad [\xi' \bar{\mathbf{W}}(\gamma \mathbf{I}_n + (1 - \gamma) \bar{\Psi})^{0.5}]^{(2)} &= [\xi' \mathbf{W}\Psi^{0.5}((1 - \gamma) \mathbf{I}_n + \gamma \Psi^{-1})^{0.5}]^{(2)} \\
 &= [\xi' \mathbf{W}((1 - \gamma) \Psi \mathbf{I}_n + \gamma \Psi \Psi^{-1})^{0.5}]^{(2)} \\
 &= [\xi' \mathbf{W}(\gamma \mathbf{I}_n + (1 - \gamma) \Psi)^{0.5}]^{(2)}.
 \end{aligned}$$

As the right-hand side of equation (A-16) is the numerator of $\mathbf{IS}_T(\theta_m, \theta_v)$, it follows that $\mathbf{IS}_T(\bar{\theta}_m, \theta_v) = \mathbf{IS}_T(\theta_m, \theta_v)$, which proves the 2nd part of Proposition 2. $\square$

Appendix B. Bootstrap

We conduct a parametric bootstrap to provide standard errors for parameter and information share estimates resulting from the 2-step estimation procedure outlined in Section II.F.1. The procedure works as follows: We first draw an independent and identically distributed (i.i.d.) sequence of random variables from a normal mixture distribution using the mixture parameters that are estimated in the 2nd (ML) step of the estimation procedure. Next, we generate simulated price series according to equation (4) using observations from the original price series as starting values, the prespecified cointegrating vectors, the 1st-step OLS estimates of the VECM parameters, and the simulated mixture residuals. The length of the simulated series equals the number of observations in the original data set plus 100. We discard the first 100 data points in order to reduce dependence on the starting values. The 2-step estimation procedure described in Section II.F.1 is then applied to the simulated data. We store the resulting parameter estimates and compute estimates of ξ according to equation (6), upper and lower bounds of the Hasbrouck (1995) information shares according to equation (8), and the $\mathbf{IS}_T$ information shares vector according to equation (15). This procedure is repeated $B = 399$ times, as suggested by Davidson and MacKinnon (2000).

Standard errors for parameter and information share estimates are computed from the empirical distribution of the bootstrap estimates. Additionally, the test statistic A based on equation (27) is calculated in each bootstrap iteration yielding its distribution under the null hypothesis that $\Sigma_u = \mathbf{W}(\gamma \mathbf{I}_n + (1 - \gamma) \Psi) \mathbf{W}'$.

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